

Project Progress Summary

Project Title

Machine Learning-Assisted Adaptive Overcurrent Relay Coordination using Random Forest and Particle Swarm Optimization on IEEE-14 Bus System

Power System

IEEE 14-bus model created in pandapower, load flow completed, six fault types simulated, 60,000-sample dataset generated.

Machine Learning

Random Forest classifier trained (~99.958% accuracy), SHAP analysis, feature importance, benchmarking, and model saved as rf_relay_model.pkl.

Optimization

Implemented relay mathematical model, objective function, constraints, particle class, and Particle Swarm Optimization (PSO).

Hybrid ML + PSO

Integrated Random Forest predictions with the adaptive PSO workflow. Current focus is validating that ML predictions meaningfully influence optimization.

Current Issue

Prediction confidence during optimization (~0.472) needs investigation by verifying feature consistency between training and inference.

Remaining Tasks

Validate across multiple fault scenarios, improve ML-PSO integration, develop Streamlit GUI, compare with conventional relay settings, and write the research paper.

Overall Progress

Approximately 85–90% complete.